Graph Theory Class Notes

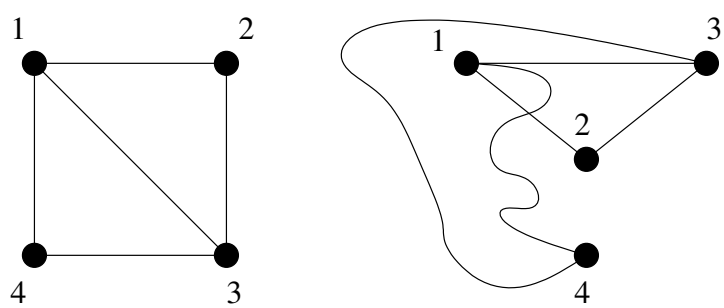
1 Introduction to Graph Theory

Definition 1.1. A graph $G = (V, E)$ consists of a non-empty set V of *vertices* and a set E of 2-element subsets of V, called *edges.*

For example

$$G = (\{1, 2, 3, 4\}, \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}, \{1, 3\}\})$$

is a graph. It has four vertices and five edges. As this mess of brackets gets troublesome to write, we drop them when it doesn't confuse things an write an edge $\{x, y\}$ as xy. The graph G can be represented pictorially as



We often write $V(G)$ for V and $E(G)$ for E if we don't define the these sets for a graph G explicitly.

There are a lot of definitions related to the fact that $e = uv$ is an edge of G. We say

- the vertices u and v are *adjacent*, written $u \sim v$,
- u and v are *neighbours*,
- u and v are the *endpoints* of e,
- u (and v) is incident with e,

Note

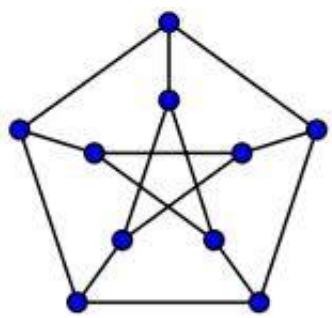
One can generalise the idea of a graph to

- *digraphs* by making edges ordered sets: uv and vu are different,
- *multigraphs* by making the edgeset E a multiset,
- *hypergraphs* by allowing edges to contain more than two vertices,
- *non-simple graphs* by allowing edges of the form uu.

We mostly don't consider these generalisations.

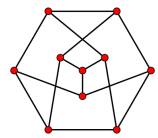
problems. Graph theory studies the properties that arise in structure due to the simple idea of a relationship between elements of a set. Many such properties are purely combinatorial, others have more structural ramifications, which arise surpisingly in most any field of mathematics.

Here is a graph, called the *Petersen Graph*, and denoted P, that some people like to put on t-shirts, and refer to fondly as the *universal counter-example*:



Here are some typical questions that we might ask of this, or any graph. The definitions you need to understand these questions come later, but they are probably what you think they are.

- How many colours do we need, when colouring the vertices of P to be sure that adjacent vertices get different colours.
- Can you draw P without intersecting edges?
- How many edges (vertices) must you remove to *disconnect* P?
- For any two pairs of vertices u, u' and v, v' can you find a *path* between u and u', and another between v and v', such that the two are disjoint?
- What is the smallest *cycle* in P?
- How many 5-cycles are there in P?

- Can we partition the edges of P into disjoint 5-cycles?
- Can we find a *walk* in P that visits every vertex (edge) exactly once?
- Is P the same graph as this: 

1.1 Isomorphisms

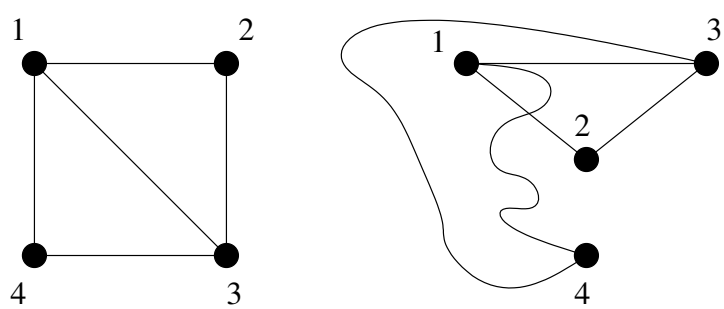
Often we don't care about the names of the vertices, and want to consider graphs to be the same if they look the same when ignoring these vertex names. For this we make the following defintion.

Definition 1.2. An *isomorphism* is a bijective map $f : V(G) \to V(H)$ such that

$$uv \in E(G) \iff f(u)f(v) \in E(H).$$

If there is an isomorphism f from G to H, we say G and H are *isomorphic*, and write $G \cong H$.

Example 1.3. The following graphs are all isomorphic.



It is a hard problem to decide if two graphs are isomorphic. We talk about this a little more later.

Theorem 1.4. *Isomorphism is an equivalence relation. That is:*

i. $G \equiv G$ *(symmetry)*

ii. $G \equiv H \Rightarrow H \equiv G$ *(reflexivity)*

iii. $G \equiv H$ *and* $H \equiv J \Rightarrow G \equiv J$ *(transitivity)*

Problem 1.1. Prove this!

Problem 1.2. How many graphs are there on 4 vertices? How many non-isomorphic (unlabelled) graphs?

As we usually consider graphs to be the same if they are isomorphic, we would like to decide if two graphs are isomorphic. Deciding if two graphs are isomorphic is usually pretty hard (this statement can be made rigourous). But if we are lucky, there are easy ways to prove that they are not.

For example, if two graphs have a different number of vertices, or a different number of edges, they are not isomorphic. Such properties, which are 'preserved by isomorphism', are called graph properties. Graph Theory is the study of these properties.

1.2 Some Common Graphs

We generally now consider graphs to be the same if they are isomorphic, and take blatent liberties with the names of their vertices. As such, we define some common graphs, but will apply these names to any graph that is isomorphic.

- The complete graph on n vertices, K_n, has $V(K_n) = \{v_1, v_2, \dots v_n\}$ and $E(K_n) = \{v_i v_j \mid 1 \leq i < j \leq n\}$.
- The n-cycle, C_n, has $V(C_n) = \{v_1, v_2, \dots v_n\}$ and $E(C_n) = \{v_i v_{i+1} \mid i = 1, \dots, n-1\} \cup \{v_n v_1\}$.
- The n-path, P_n, has $V(P_n) = \{v_0, v_1, \dots v_n\}$ and $E(P_n) = \{v_i v_{i+1} \mid i = 0, \dots, n-1\}$.
- The complete bipartite graph, $K_{m,n}$, has $V(K_{m,n}) = \{a_1, \dots a_m\} \cup \{b_1, \dots, b_n\}$ and $E(K_{m,n}) = \{a_i b_j \mid i \in [m], j \in [n]\}$.
- The n-cube, Q_n, $V(Q_n)$ is the set of binary strings of length n, vertices u and v are adjacent if they differ in exactly one coordinate.

1.3 Homomorphisms

Definition 1.5. A *homomorphism* $f : G \to H$ of a graph G to another graph H, is a vertex map $f : V(G) \to V(H)$ such that

$$uv \in E(G) \Rightarrow f(u)f(v) \in E(H).$$

If there is a homomorphism $f : G \to H$, we write simply $G \to H$.

Any isomorphism is a homomorphism, and as usual, a homomorphism $f : G \to H$ is an isomorphism if and only if there is an inverse homomorphism $f^{-1} : H \to G$ such that $f \circ f^{-1}$ and $f^{-1} \circ f$ are the identity maps. But there is another property you may be used to for other finite structures, that doesn't hold for us: **a bijective homomorphism is not necessarily an isomorphism.** Indeed, there are many bijections that are not isomorphisms.

Example 1.6. For any graph G on n vertices there is a bijective homomorphism $f : G \to K_n$.

A homomorphism $f : G \to H$ is a map on the vertices, but it induces a map $f_E : E(G) \to E(H)$ on the edges:

$$f_E(uv) = f(u)f(v).$$

The homomorphism f is called *edge-injective* if f_E is injective, and *edge-surjective* if f_E is surjective.

Problem 1.3. Show that if f is injective, then it is edge-injective, but that the converse is not true.

Problem 1.4. Show that f is an isomorphism if and only if it is bijective and edge-surjective.

Problem 1.5. Show that $C_{2n+1} \to C_{2m+1}$ if and only if $m \leq n$. Show that $C_n \to K_2$ if and only if n is even. Show that $G \to K_2 \Rightarrow C_{2n+1} \not\to G$ for every n.

1.4 Subgraphs and Complements

Given a graph G, another graph G' is a *subgraph* of G, written $G' \leq G$ if $V(G') \subset V(G)$ and $E(G') \subset E(G)$.

A subgraph G' of G is *spanning* if $V(G') = V(G)$, or *induced* if $E(G') = \{uv \in E(G) | u, v \in V(G')\}$. A subgraph G' of G is a *proper* subgraph of G if $E(G')$ is a proper subset of $E(G)$.

Problem 1.6. How many non-labelled subgraphs are there of K_4? How many are proper? ...induced? ...spanning? ...spanning, induced and proper?

Definition 1.7. The *complement* $\overline{G}$ of a graph G is the graph defined by $V(\overline{G}) = V(G)$ and $E(\overline{G}) = \overline{E(G)}$, (where the universe used for complementation is the set $\{uv \mid u \neq v \in V(G)\}$ of all possible edges on $V(G)$).

Problem 1.7. Show that a subgraph G' of G is induced if and only if $\overline{G'}$ is a subgraph of $\overline{G}$.

Solution

Let $G' \subset G$ be induced. Clearly $V(\overline{G'}) \subset V(\overline{G})$, so we have to show that $E(\overline{G'}) \subset E(\overline{G})$. Let $e \in E(\overline{G'})$ then $e \notin E(G')$ so $e \notin E(G)$. So $e \in E(\overline{G})$.
On the other hand, let $\overline{G'} \leq \overline{G}$. We want to show that G' is induced. For $u, v \in V(G')$, with $uv \in E(G)$ we want to show that $uv \in E(G')$. Well $uv \in E(G)$ implies that $uv \notin E(\overline{G})$ so not in $E(\overline{G'})$, so because $u, v \in V(G')$, $uv \in E(G)$.

Problem 1.8. Show that a bijective homomorphism $f : G \to H$ is an isomorphism if and only if it is also a homomorphism of $\overline{G} \to \overline{H}$.

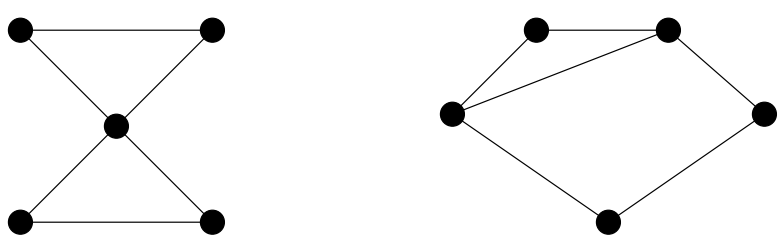
Problem 1.9. A graph G is self-complementary if $G \equiv \overline{G}$. How many edges are there in a self-complementary graph on n vertices?

1.5 Degrees and Degree Sequences

Definition 1.8. For a vertex v in a graph G, the *degree* of v, written $\deg(v)$, is the number of edges in G incident to v. The *degree sequence* of a graph G is the multiset of the degrees of the vertices of G. The minimum degree, $\delta(G)$, and maximum degree, $\Delta(G)$, of a graph G are the minimum and maximum degrees respectively in its degree sequence.

Problem 1.10. Show that every graph contains a path of length at least $\delta(G)$.

Clearly, degree sequence, minimum degree, and maximum degrees are graph properties. So the following two graphs, which have the same number of vertices, and edges are not isomorphic as they have different degree sequences: $(4, 2, 2, 2, 2)$ and $(3, 3, 2, 2, 2)$ respectively, (and in fact different maximum degrees.)



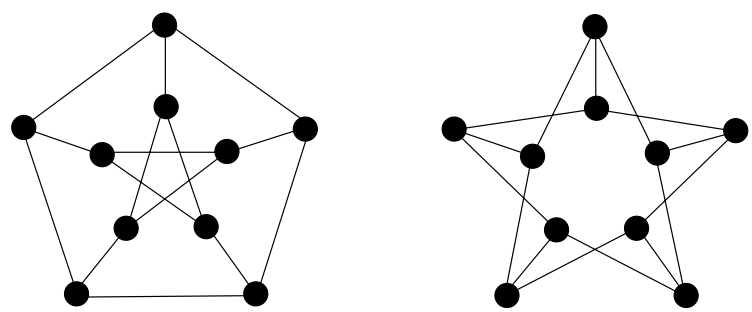
Here is a simple observation about degree sequence:

Theorem 1.9. *If $(d_1, d_2, \ldots, d_n)$ is the degree sequence of a graph, then the degree sum: $\sum_{i=1}^{n} d_i$ is equal to $2|E(G)|$.*

Proof. Each edge of the graph contributes 2 to the degree sum, one for each endpoint. □

Corollary 1.10. *In any graph G, there is an even number of vertices of odd degree.*

Problem 1.11. What are the degree sequences of the following graphs? Are the following two graphs, with the same degree sequences, isomorphic?



Problem 1.12. What is the degree sequence of the cube Q_n?

Definition 1.11. A graph G is called *k-regular* if every vertex has degree k.

Problem 1.13. How many edges does a k-regular graph on n vertices have?

A sequence $D = (d_1, d_2, \ldots, d_n)$ non-increasing integers is *graphic* if it is the degree sequence of a graph G. Such a graph G is called a realisation of the D. We saw above that one necessary condition for a sequence to be graphic is that it sums to an even number. But this is not sufficient: $(4, 2, 0)$ is not graphic.

Problem 1.14. Show that the following condition is also necessary for a sequence $(d_1, d_2, \ldots, d_n)$ to be graphic. For all $1 \leq k \leq n$,

$$\sum_{i=1}^{k} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(k, d_i).$$

In fact, this condition is also sufficient. But this is fair bit harder to show:

Theorem 1.12. *(Erdős-Gallai '60) A sequence $(d_1, d_2, \ldots, d_n)$ is graphic if and only if $\sum_{i=1}^{k} d_i$ is even and for all $1 \leq k \leq n$,*

$$\sum_{i=1}^{n} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(k, d_i).$$

Note

It isn't so hard to prove. Look up a 2010 paper by Tripathi, Venugopalan and West for a nice short proof.

The Erdős-Gallai condition is what we use to decide if a sequence is graphic, but there is also a recursive method to decide it. It depends on the following idea:

Lemma 1.13. *(Havel-Hakimi) If a sequence $D = (d_1, d_2, \ldots, d_n)$ is graphic, then there is a graph with degree seqence D such that the maximum degree vertex is adjacent to the d_1 vertices of next highest degree.*

Thus to decide if a sequence D, such as $(3, 3, 2, 2, 2)$, is graphic, the *Havel-Hakimi algorithm* is as follows. While D is not all zeros,

- remove d_1, and
- remove 1 from the next d_1 degrees $d_2, \ldots, d_{d_1-1}$,

if possible.

The algorithm clearly terminates. If it terminates with an empty sequence, (or a sequence of zeros), then D is graphic. Otherwise, it is not graphic.

Starting with $(3, 3, 2, 2, 2)$, we get $(\cancel{3}, 3-1, 2-1, 2-1, 2) = (2, 2, 1, 1)$, and then $(1, 0, 1) = (1, 1)$, and then $(0, 0)$. So $(3, 3, 2, 2, 2)$ is graphic.

After one definition, let's prove the Lemma.

Definition 1.14. The *neighbourhood* $N(v)$ of a vertex v in a graph G is the set

$$\{u \in V(G) \mid uv \in E(G)\}$$

of all neighbours of v.

Proof. Let $D = (d_1, d_2, \ldots, d_n)$ be a graphic sequence. Given a realisation G of D, let v be a vertex of degree $\Delta = d_1$, and let $T = (v_2, \ldots, v_{\Delta+1})$ be a set of Δ vertices where v_i has degree d_i for all i. If $N(v) = T$ then we are done, so towards contradiction, assume that the realisation G of D that maximises $\Delta' = |N(v) \cap T|$ has $\Delta' < \Delta$.

We find a new realisation G' with $|N(v) \cap T| > \Delta'$, so the lemma follows by contradiction.

As $N(v) \neq T$, there is $t \in T$ with $v \not\sim t$ and $s \notin T$ with $v \sim s$. Furthermore, since $t \in T$ and $s \notin T \cup \{v\}$, $\deg(t) \geq \deg(s)$. So there is some vertex w with $w \sim t$ and $w \not\sim s$.

Removing the edges vs and tw from G and adding vt and sw we get a graph G' with the same degrees on every vertex, (so another realisation of D), but with $|N_{G'}(v) \cap T| = |(N_G(v) \cap T) \cup \{t\}| = \Delta' + 1$, as needed for the contradiction. $\square$

Problem 1.15. Decide, using Havel-Hakimi, whether or not the sequence $(5, 5, 5, 3, 2, 2, 1, 1)$ is graphic.